

SOLUTION QUIZ-03 BSE-6B

QUESTION-01

a.

$$D_{trans} = \frac{L}{R} = \frac{4000}{100,000,000} = 0.00004 \text{ seconds} = 0.04 \text{ ms}$$

$$U_{sender} = \frac{D_{trans}}{D_{trans} + RTT}$$

$$U_{sender} = \frac{0.04}{0.04 + 10} = \frac{0.04}{10.04} \approx 0.003984$$

b.

To achieve 100% utilization, the sender must continuously transmit packets during the entire time it takes to get the first acknowledgment back (which is one full round trip plus the transmission time of the first packet).

We can calculate the required window size (W) by dividing the total time of one cycle by the transmission time of a single packet:

$$W = \frac{D_{trans} + RTT}{D_{trans}}$$

$$W = \frac{10.04 \text{ ms}}{0.04 \text{ ms}} = 251$$

Answer: A minimum window size of **251 packets** must be kept in flight to achieve 100% utilization.

QUESTION-02

Total number of packets (including both transmissions and re-transmissions) sent by host A is 16 as shown in the following timing diagram:

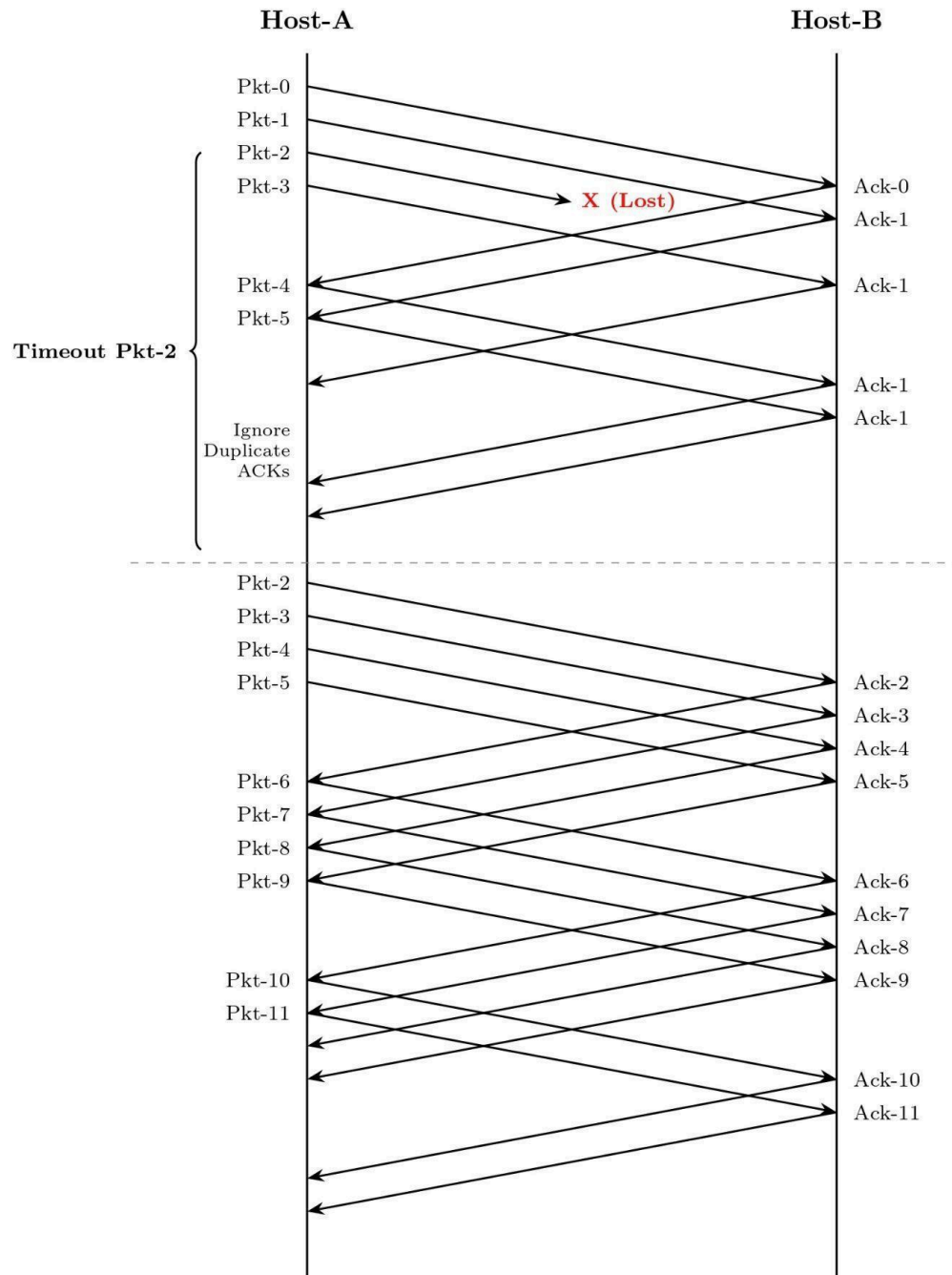


Figure 1: Go-Back-N Protocol Timing Diagram